# Jetson Operation

We will use the existing Communication Protocol between the Lift Controller and AMR, but with slight adjustments:

1. RS485/RS422 connection
2. 38400 bps for Lift IO Protocol with 8-bit data, no parity and 1 stop bit
3. Jetson is the initiator, and the LC is the responder
4. Jetson side to poll lift controller sequentially. Lift controller to reply within 100ms

# Function Codes

The Jetson-to-LC protocol reuses the existing AMR-LC frame format, including the 0xA5A5 header, Data Length field, Lift ID field, Function Code field, Data Portion field, and 2-byte CRC.

Function codes 0x10, 0x11, and 0x12 are defined for Jetson-LC communication. Each new function code has its own Data Portion size and byte/bit mapping.

Data Length refers to the number of bytes from Lift ID to the end of the Data Portion. It excludes the Header and CRC bytes.

All 2-byte numerical values are transmitted in high-byte-first order.

## Function Code 0x10

Used to request lift configuration data and Distance from Car Top to Current Floor Sill from the LC.

1. From Jetson to LC:

```
Header – 0xA5A5
Data Length – 0x03
Lift ID – 0x01 ~ 0x0n
Function Code – 0x10
Data Portion – 0x00
CRC – 2 CRC bytes
```

2. From LC to Jetson (Reply)

```
Header – 0xA5A5
Data Length – 0x22
Lift ID – 0x01 ~ 0x0n
Function Code – 0x10
```

```
Data Portion – 32 bytes (See Table 1)
CRC – 2 CRC bytes
```

## Function Code 0x11

Used to retrieve specific operational status bits and automatic service session status from the LC.

1. From Jetson to LC:

```
Header – 0xA5A5
Data Length – 0x03
Lift ID – 0x01 ~ 0x0n
Function Code – 0x11
Data Portion – 0x00
CRC – 2 CRC bytes
```

2. From LC to Jetson (Reply)

```
Header – 0xA5A5
Data Length – 0x06
Lift ID – 0x01 ~ 0x0n
Function Code – 0x11
Data Portion – 4 bytes (See Table 2)
CRC – 2 CRC bytes
```

## Function Code 0x12

Used to send commands to the lift.
0x12 is also used by the Jetson to start/hold the robot maintenance request, clear the robot maintenance request when servicing is complete, or abort the robot maintenance operation.

1. From Jetson to LC:

```
Header – 0xA5A5
Data Length – 0x05
Lift ID – 0x01 ~ 0x0n
Function Code – 0x12
Data Portion – 3 bytes (See Table 5)
CRC – 2 CRC bytes
```

2. From LC to Jetson (Reply)

```
Header – 0xA5A5
Data Length – 0x03
Lift ID – 0x01 ~ 0x0n
```

```
Function Code – 0x12
Data Portion – 0x00
CRC – 2 CRC bytes
```

For `0x12` request changes, the Jetson shall resend the same command until either:

1. the LC acknowledges the command, or
2. the Jetson observes the expected state transition through `0x11` status polling.

Once the request change is acknowledged or the expected state transition is observed, the Jetson shall stop retrying that request change.

The `0x12` reply only confirms that the command frame was received and accepted by the LC. The Jetson shall monitor the actual progress and result of the automatic service session using `0x11`.

While robot maintenance operation remains active, the Jetson shall continue sending periodic `0x12` hold frames with:

Automatic Service Request = 1 (ON)
Automatic Service Request Type = 0x01

Repeated `0x12` frames with the same held request state shall be treated as the same active robot maintenance request, not as repeated new start requests.

## Table 1: `0x10` Data Portion

Example Data Portion:

- Rated Capacity: 900kg
- Car Width: 1600mm
- Distance from Car Top to Current Floor Sill: 1400mm
- Door Width: 1000mm
- 2 Doors: No
- 2nd Door Position: NIL
- Service Floors: 16 Floors with 5th Floor Non-Service

Note: Service Floor Config Bit 0 is the LSB.

Note: The Distance from Car Top to Current Floor Sill value is derived from LC-side lift position information and may be affected by motor encoder error, rope slip, lift leveling issues, power failure, encoder reset, or recalibration requirements. Therefore, this value should be treated as a coarse positioning reference or sanity check, rather than the only final positioning reference for robot inspection.

For final inspection positioning, the Jetson may also use the current floor information from `0x11` together with camera-based detection of a defined inspection point for each service floor.

| | Byte No. | Data | Description |
|---|---|---|---|
| **Rated Capacity** | **0** | **03** | **900 Decimal = 0384 Hexadecimal** |
| | **1** | **84** | |
| **Car Width** | **2** | **06** | **1600 Decimal = 0640 Hexadecimal** |
| | **3** | **40** | |
| **Distance from Car Top to Current Floor Sill** | **4** | **05** | **1400 Decimal = 0578 Hexadecimal** |
| | **5** | **78** | |
| **Door Width** | **6** | **03** | **1000 Decimal = 03E8 Hexadecimal** |
| | **7** | **E8** | |
| **2 Doors** | **8** | **00** | **No second door** |
| **2nd Door Position** | **9** | **00** | **No second door position** |

| | Byte No. | Data | Description |
|---|---|---|---|
| **Service Floor Config** | **10** | **EF** | **1110 1111 (Floor 5 is out of service)** |
| | **11** | **FF** | **1111 1111 (All floors are working)** |
| | **12** | **00** | |
| | **13** | **00** | |
| | **14** | **00** | |
| | **15** | **00** | |
| | **16** | **00** | |
| | **17** | **00** | |
| | **18** | **00** | |
| | **19** | **00** | |
| | **20** | **00** | |
| | **21** | **00** | |
| | **22** | **00** | |
| | **23** | **00** | |
| | **24** | **00** | |
| | **25** | **00** | |
| **Spare** | **26** | **00** | |
| **Spare** | **27** | **00** | |
| **Spare** | **28** | **00** | |
| **Spare** | **29** | **00** | |
| **Spare** | **30** | **00** | |
| **Spare** | **31** | **00** | |

## Table 2: `0x11` Data Portion

For this Jetson maintenance design, the Jetson shall only operate the robot or perform Manual Down positioning when the LC reports that Jetson operation is permitted, Robot Maintenance Mode is ON, and Lift Abnormal Mode is OFF.

| | | | |
|---|---|---|---|
| **Byte 0** | | **Lift Current Floor** | **1 as lowest floor** |
| **Byte 1** | **Bit 0** | **Jetson Operation Permitted** | **0 = OFF; 1 = ON** |
| | **Bit 1** | **Lift arrived at destination floor** | **0 = OFF; 1 = ON** |
| | **Bit 2** | **Reserved / Future Door Status** | **0 = OFF; 1 = ON** |
| | **Bit 3** | **Reserved / Future Door Status** | **0 = OFF; 1 = ON** |
| | **Bit 4** | **Robot Maintenance Mode** | **0 = OFF; 1 = ON** |
| | **Bit 5** | **Lift Abnormal Mode** | **0 = OFF; 1 = ON** |
| | **Bit 6** | **Spare** | **0 = OFF; 1 = ON** |
| | **Bit 7** | **Spare** | **0 = OFF; 1 = ON** |
| **Byte 2** | | **Automatic Service Session State** | **See Table 3** |
| **Byte 3** | | **Automatic Service Abnormal Reason** | **See Table 4** |
| **Robot Maintenance Mode → LC-controlled robot maintenance operation<br>Lift Abnormal Mode → Lift under any other state that Jetson cannot operate on** | | | |

| Byte 2 Value | Automatic Service Session State | Description |
|---|---|---|
| `0x00` | **Idle** | **No automatic service session is active.** |
| `0x01` | **Preparation Request Received** | **LC has received the automatic service preparation request.** |
| `0x02` | **Checking Readiness** | **LC is checking whether the lift is available for automatic service.** |
| `0x03` | **Not Available** | **Lift is not available for automatic service.** |
| `0x04` | **Entering Robot Maintenance Mode** | **LC is changing the lift into Robot Maintenance Mode and disabling normal passenger operation.** |
| `0x05` | **Moving to Top Service Floor** | **LC is moving the lift to the top service floor.** |
| `0x06` | **Ready for Jetson/Robot Control** | **LC has completed automatic service preparation and the lift is ready for Jetson-controlled robot operation.** |
| `0x07` | **Completion Request Received** | **LC has detected that Automatic Service Request was cleared to OFF after an active robot maintenance operation and has accepted the return-to-normal request.** |
| `0x08` | **Returning to Auto Mode** | **LC is restoring Auto mode and normal passenger operation.** |
| `0x09` | **Service Complete / Normal Operation Restored** | **LC has completed the automatic service session and normal operation has been restored.** |
| `0x0A` | **Failed** | **Automatic service session has failed. Refer to Automatic Service Abnormal Reason.** |
| `0x0B` | **Aborted** | **Automatic service session has been aborted.** |

Some intermediate states, such as Checking Readiness, Entering Robot Maintenance Mode, and Returning to Auto Mode, may occur almost instantly. They are retained in the protocol for visibility, diagnostics, and debugging.

## Note

`Ready for Jetson/Robot Control` means:

```
- LC has completed automatic service preparation
- Lift is in Robot Maintenance mode
- Hall calls are disabled
- Lift is stationary when control is handed over to the Jetson
- Doors are closed / no door movement is active
- Jetson Operation Permitted is ON
```

```
- Robot Maintenance Mode is ON
- Lift Abnormal Mode is OFF
- Jetson is allowed to perform manual down positioning and robot inspection
```

## Table 4: Automatic Service Abnormal Reason

| Byte 3 Value | Automatic Service Abnormal Reason | Description |
|---|---|---|
| `0x00` | No Abnormal Condition | No automatic service abnormal condition is present. |
| `0x01` | Lift Not Available | LC determined that the lift is not available for automatic service. |
| `0x02` | Door Not Closed | Door condition is not suitable for automatic service preparation or completion. |
| `0x03` | Passenger / Service Conflict | Lift cannot be taken over due to passenger operation, service demand, or other conflict. |
| `0x04` | Lift Abnormal Mode Detected | LC detected that the lift entered a mode or condition where Jetson/robot operation is not permitted, such as fire operation, emergency operation, manual takeover, technician operation, or other LC-side override mode. |
| `0x05` | Failed to Enter Robot Maintenance Mode | LC failed to place the lift into Robot Maintenance mode. |
| `0x06` | Failed to Return to Auto Mode | LC failed to restore Auto mode or normal operation. |
| `0x07` | Timeout | LC automatic service session exceeded the expected allowed time. |
| `0x08` | Safety / Interlock Condition | LC detected a safety-chain, door interlock, lift fault, or other safety-related condition preventing the session from continuing. |
| `0x09` | Internal LC Error | LC encountered an internal error while processing the automatic service session. |
| `0x0A` | Manual Technician Intervention / Manual Service Takeover | LC detected a manual technician intervention or manual service takeover during the automatic service session. |
| `0x0B` | Abort Requested by Jetson | Jetson requested the LC to abort the automatic service session. |
| `0x0C` | Jetson Communication Lost | LC detected that Jetson communication was lost during the automatic service session. |

Note: The `Timeout` abnormal reason may include cases where the automatic servicing workflow exceeds the configured maximum servicing duration. The initial proposed

maximum duration for the full servicing process is `1 hour`, measured from the `0x12` Start Automatic Service Preparation request until `Service Complete / Normal Operation Restored` is reported.

If the maximum servicing duration is exceeded, the Jetson shall record the automatic service session result in its logs, clear Manual Up and Manual Down commands if active, and clear Automatic Service Request to OFF to request LC return-to-normal handling, unless prevented by LC-side safety logic.

## Table 5: `0x12` Data Portion

When no destination floor command is required, Byte 0 shall be set to `0x00`, and the LC shall ignore Byte 0 for non-destination commands.

When `Automatic Service Request = 1`, Byte 2 shall determine the automatic service request type, and the LC shall ignore Byte 0 unless required by LC-side logic.

| **Byte 0** | | **Destination Floor** | **1 as lowest floor, MSB denotes 2nd door** |
|---|---|---|---|
| **Byte 1** | **Bit 0** | **Reserved / Future Door Command** | **0 = OFF; 1 = ON** |
| | **Bit 1** | **Reserved / Future Door Command** | **0 = OFF; 1 = ON** |
| | **Bit 2** | **Hall call disable command** | **0 = OFF; 1 = ON** |
| | **Bit 3** | **Manual up command** | **0 = OFF; 1 = ON** |
| | **Bit 4** | **Manual down command** | **0 = OFF; 1 = ON** |
| | **Bit 5** | **Automatic Service Request** | **0 = OFF; 1 = ON** |
| | **Bit 6** | **Spare** | |
| | **Bit 7** | **Spare** | |
| **Byte 2** | | **Automatic Service Request Type** | **See Table 6** |

### Note

Door command bits are reserved for possible future robot maintenance use. In the current robot maintenance design, the Jetson shall keep these bits OFF and the LC shall ignore them.

Automatic Service Request shall be treated as a held robot maintenance request.

When Automatic Service Request = 0, Automatic Service Request Type shall be set to 0x00. The LC shall treat this as no active robot maintenance request, or as a request-cleared / return-to-normal indication if a robot maintenance operation was previously active.

The Jetson shall keep Automatic Service Request = ON while robot maintenance operation is active. This includes LC preparation, Jetson-controlled manual down positioning, robot inspection, and LC completion handling.

When robot maintenance operation is completed, the Jetson shall clear Automatic Service Request = OFF.

The LC shall use this held request as a continuous indication that Jetson/robot maintenance operation is still active.

Repeated 0x12 frames with the same Automatic Service Request state shall be treated as the same held request, not as repeated new service requests.

During Jetson-controlled manual down positioning, the command shall be sent with:

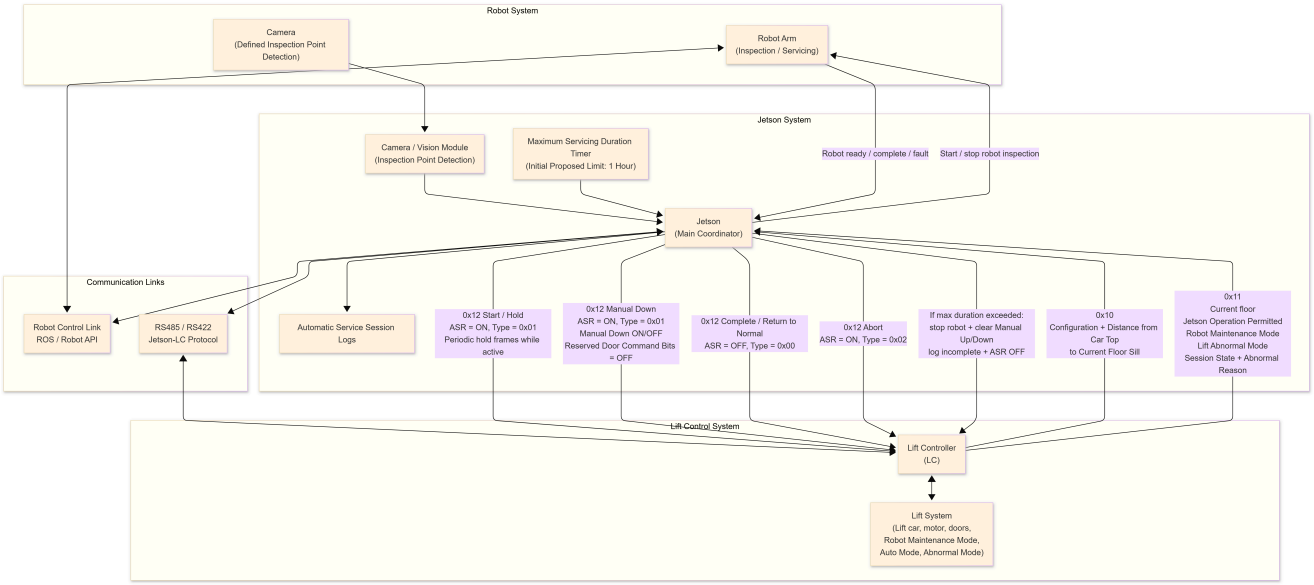
```
Automatic Service Request = 1 (ON)
Automatic Service Request Type = 0x01
Request Type Meaning = Start/Hold Robot Maintenance Operation
Hall Call Disable = 1 (ON)
Manual Down = 1 (ON) when moving downward
Manual Down = 0 (OFF) when stopping
Reserved Door Command Bits = 0 (OFF)
```

## Table 6: Automatic Service Request Type

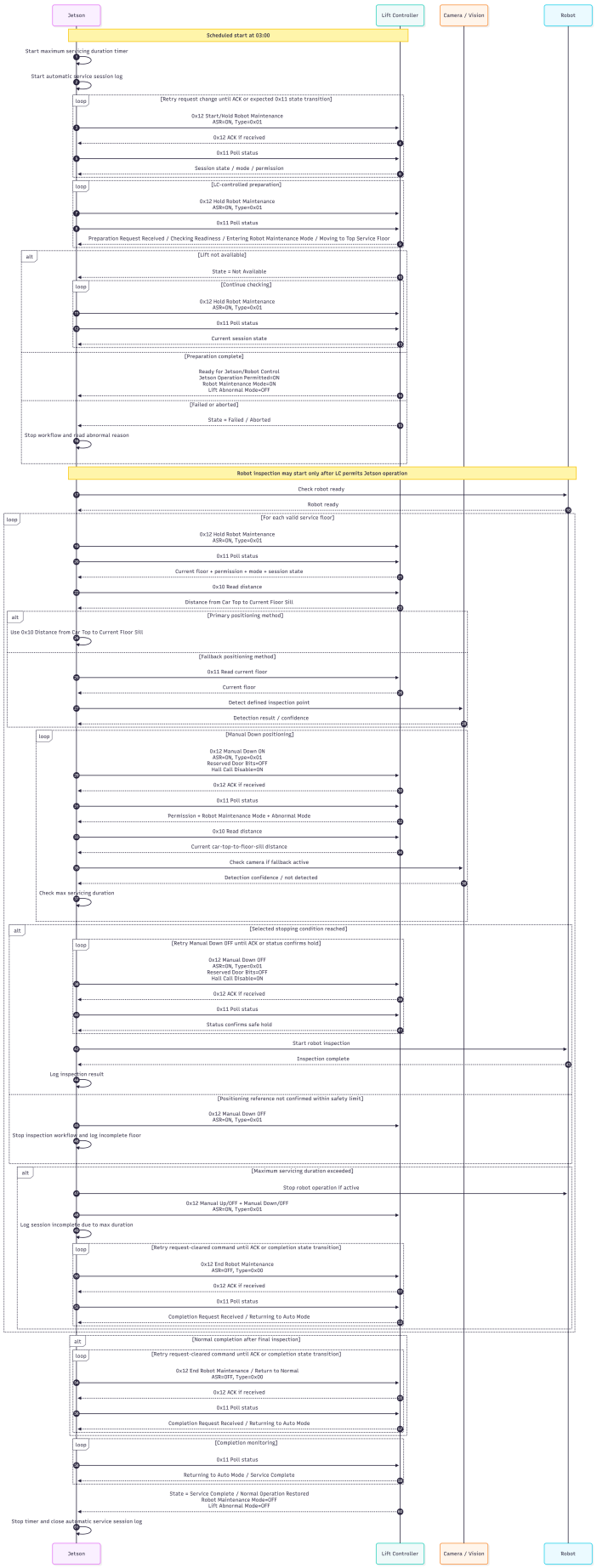
| Byte 2 Value | Automatic Service Request Type | Description |
|---|---|---|
| `0x00` | No Robot Maintenance Request | No robot maintenance request is active. When sent after an active held request, this indicates that robot maintenance operation is complete and the LC may begin return-to-normal handling. |
| `0x01` | Start/Hold Robot Maintenance Operation | LC checks readiness, disables hall calls, enters Robot Maintenance Mode, moves/homes the lift to the top service floor, and reports Ready for Jetson/Robot Control. While Automatic Service Request remains ON, LC treats robot maintenance operation as still active. |
| `0x02` | Abort Robot Maintenance Operation | LC aborts the robot maintenance session and moves to a safe stopped/held state according to LC safety logic. |

# Diagrams

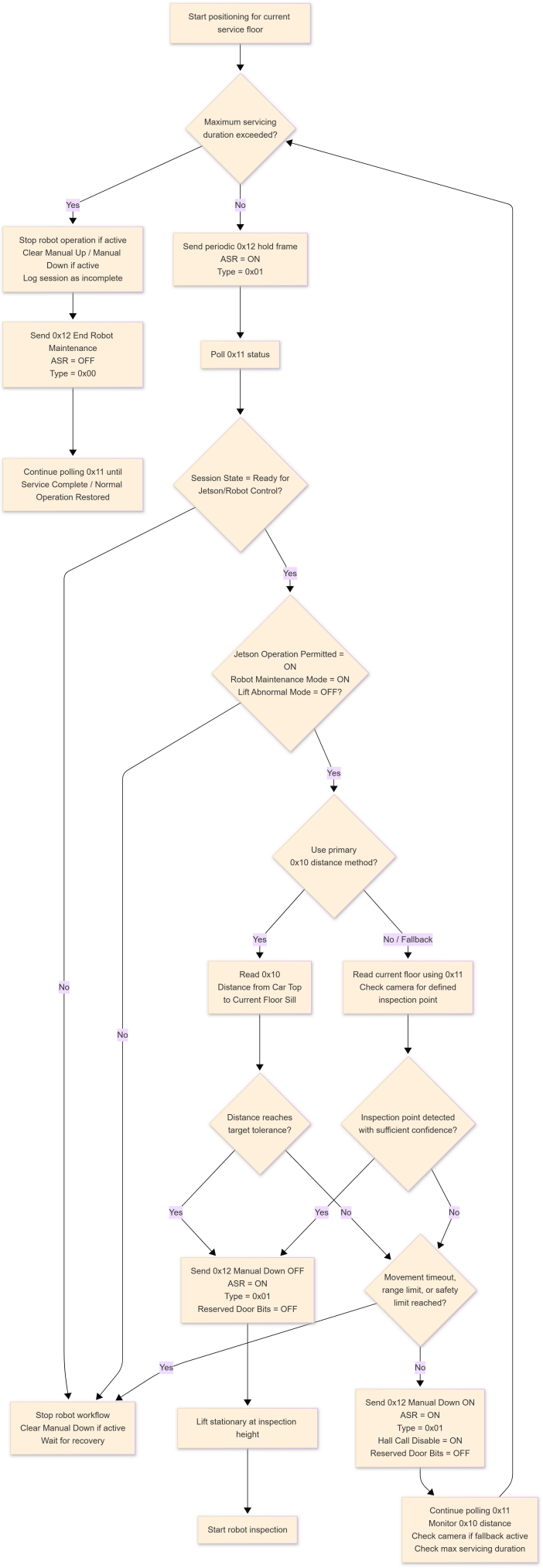
## System Architecture Diagram



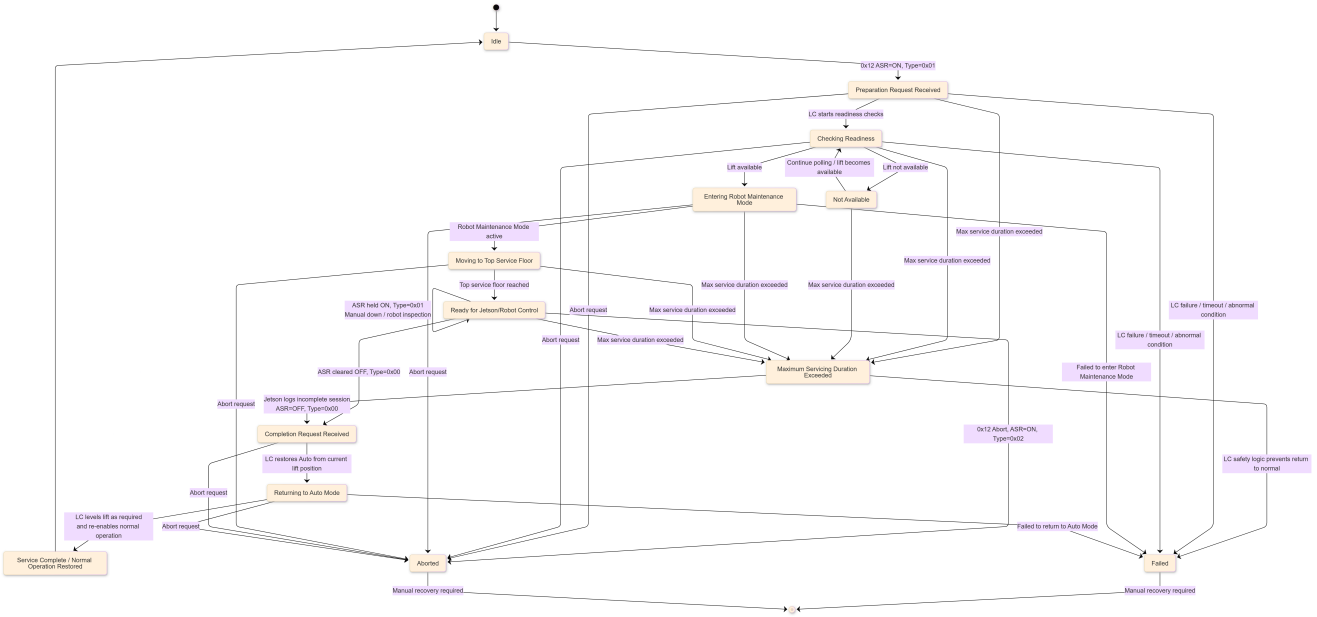
# Main Maintenance Sequence Diagram



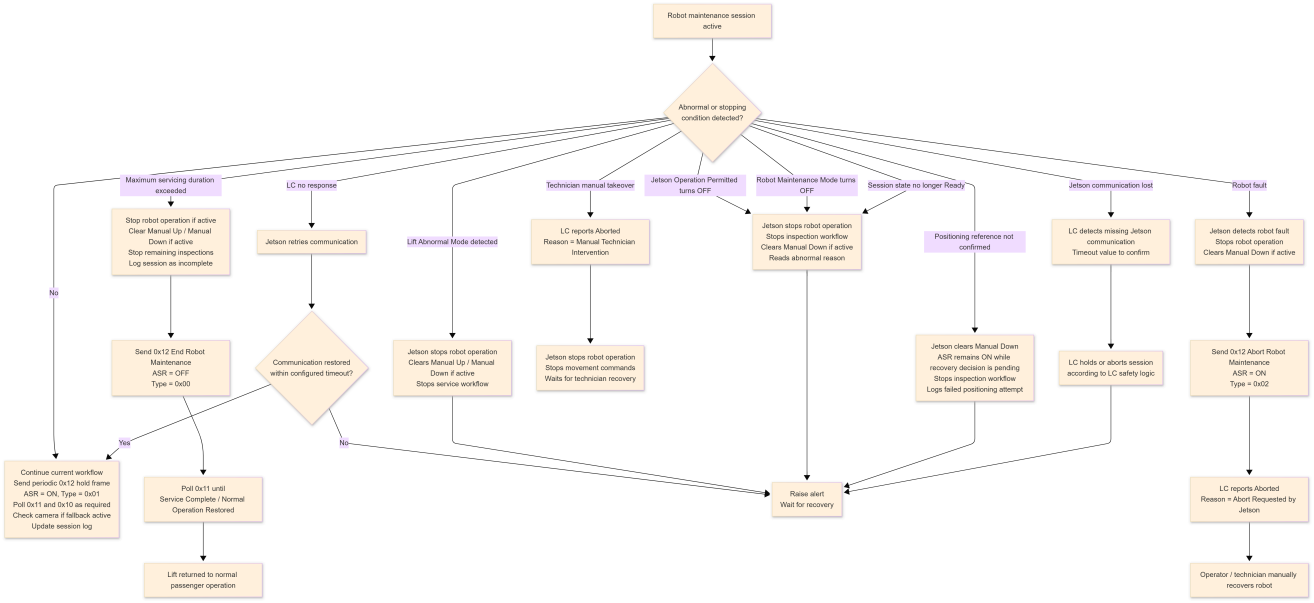

# Manual Down Positioning Loop Diagram


Start positioning for current service floor
Maximum servicing duration exceeded?
Yes
No
Stop robot operation if active
Clear Manual Up / Manual Down if active
Log session as incomplete
Send periodic 0x12 hold frame
ASR = ON
Type = 0x01
Send 0x12 End Robot Maintenance
ASR = OFF
Type = 0x00
Poll 0x11 status
Continue polling 0x11 until Service Complete / Normal Operation Restored
Session State = Ready for Jetson/Robot Control?
Yes
No
Jetson Operation Permitted = ON
Robot Maintenance Mode = ON
Lift Abnormal Mode = OFF?
Yes
No
Use primary 0x10 distance method?
Yes
No / Fallback
Read 0x10 Distance from Car Top to Current Floor Sill
Read current floor using 0x11 Check camera for defined inspection point
Distance reaches target tolerance?
Inspection point detected with sufficient confidence?
Yes
Yes
No
No
Send 0x12 Manual Down OFF
ASR = ON
Type = 0x01
Reserved Door Bits = OFF
Movement timeout, range limit, or safety limit reached?
Yes
No
Stop robot workflow
Clear Manual Down if active
Wait for recovery
Lift stationary at inspection height
Send 0x12 Manual Down ON
ASR = ON
Type = 0x01
Hall Call Disable = ON
Reserved Door Bits = OFF
Start robot inspection
Continue polling 0x11
Monitor 0x10 distance
Check camera if fallback active
Check max servicing duration

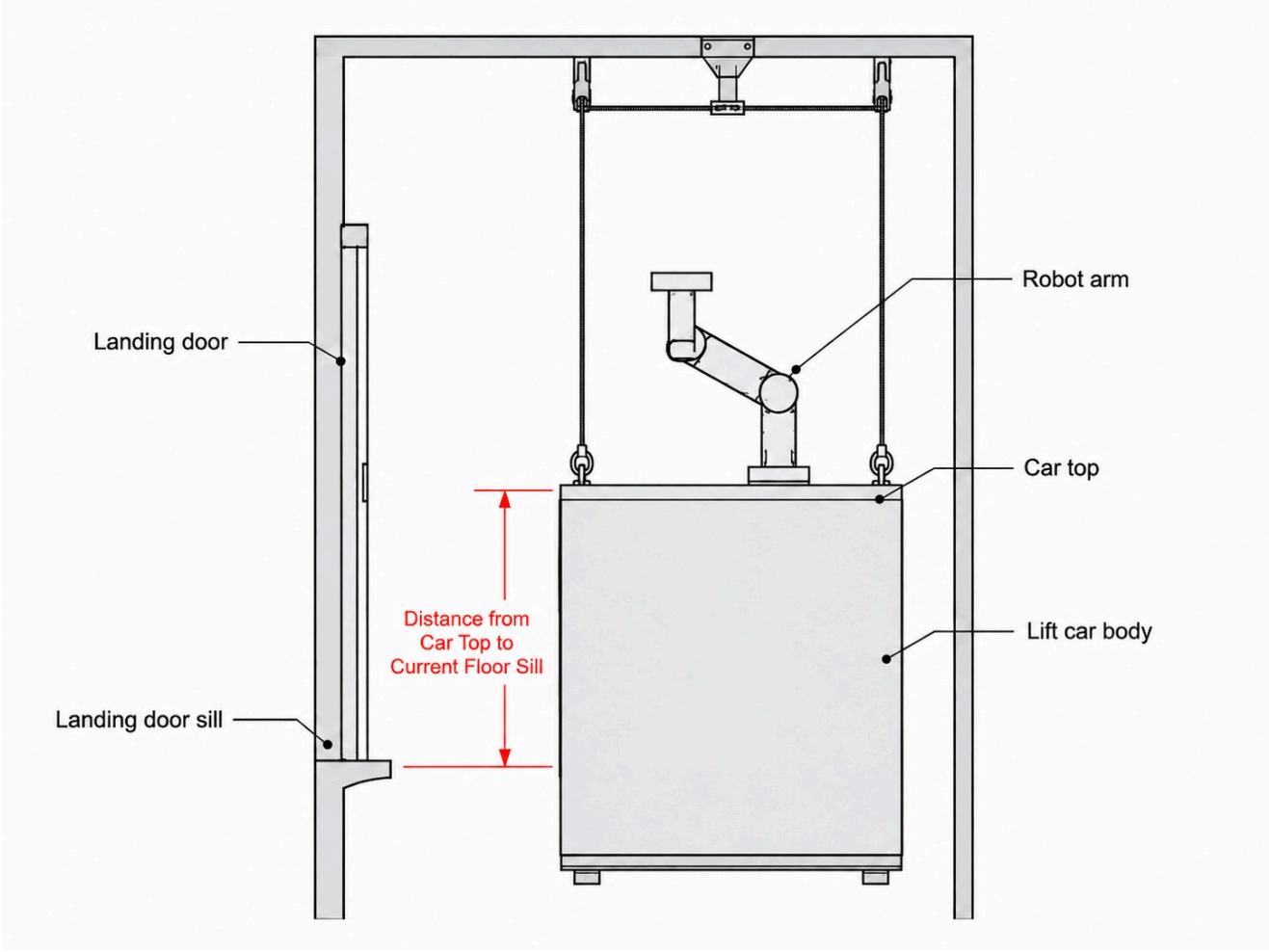
# Automatic Service Session State Machine Diagram



# Abnormal / Override Handling Diagram




# Distance from Car Top to Current Floor Sill Illustration



# Testing Plan

The system should be tested in phases so that each integration layer can be verified independently before testing the full nightly maintenance workflow. This reduces debugging complexity and allows the Jetson-LC protocol, LC automatic service handling, robot control, manual down positioning, and abnormal handling to be validated step by step.

The testing plan follows the current design:

```
- LC handles automatic service preparation.
- LC handles automatic service completion.
- Jetson polls LC status using 0x11.
- Jetson handles manual down positioning.
- Jetson controls the robot inspection workflow.
- LC reports Automatic Service Session State and Automatic Service Abnormal
Reason.
```

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# Phase 1: Jetson-LC Protocol Integration Test

## Objective

Verify that the Jetson can communicate reliably with the Lift Controller using the proposed RS485/RS422 protocol.

## Scope

This phase focuses only on communication between the Jetson and the LC. The robot is not involved.

## Test Items

- Verify that the Jetson can send valid `0x10`, `0x11`, and `0x12` frames.
- Verify that the LC replies within the required `100 ms` response time.
- Verify correct CRC generation and CRC validation.
- Verify correct Lift ID handling.
- Verify correct parsing of the Data Length field.
- Verify that `0x10` returns lift configuration data and Distance from Car Top to Current Floor Sill.
- Verify that `0x11` returns the 4-byte status data portion.
- Verify that `0x11` correctly reports:
  - Current floor
  - Jetson Operation Permitted

    - Robot Maintenance Mode
    - Lift Abnormal Mode
    - Automatic Service Session State
    - Automatic Service Abnormal Reason
- Verify that `0x12` command frames are accepted and acknowledged by the LC.
- Verify that `0x12` request changes are retried until ACK or the expected `0x11` state transition is observed.
- Verify that periodic `0x12` hold frames can be sent while robot maintenance operation is active.
- Verify that repeated `0x12` hold frames with `Automatic Service Request = 1` and `Automatic Service Request Type = 0x01` are treated as the same active robot maintenance request, not repeated new start requests.
- Verify that `Automatic Service Request = 0` and `Automatic Service Request Type = 0x00` is interpreted as no active robot maintenance request, or as request-cleared / return-to-normal if a robot maintenance operation was previously active.
- Verify that `Automatic Service Request = 1` and `Automatic Service Request Type = 0x02` is interpreted as Abort Robot Maintenance Operation.
- Verify that repeated `0x12` frames with the same held request state are treated as the same active robot maintenance request, not repeated new start requests.
- Verify timeout and retry handling when the LC does not respond.

## Expected Result

The Jetson and LC can exchange valid protocol frames reliably. The Jetson can correctly read configuration data, status data, session state, abnormal reason, and send accepted command frames to the LC.

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# Phase 2: LC Automatic Service Preparation Test

## Objective

Verify that the LC can handle automatic service preparation after receiving the Jetson's start request.

## Scope

This phase focuses only on the Jetson-LC preparation flow. The robot is not involved.

## Test Items

- Verify that the Jetson can send `0x12` with:

```
Automatic Service Request = 1
Automatic Service Request Type = 0x01
```

- Verify that the LC acknowledges the `0x12` command.
- Verify that the Jetson does not send Manual Up, Manual Down, destination floor, or door commands while LC preparation is in progress.
- Verify that the Jetson continues sending periodic `0x12` hold frames with `Automatic Service Request = 1` and `Automatic Service Request Type = 0x01`, while also polling `0x11` during LC preparation.
- Verify that the Jetson does not send Manual Up, Manual Down, destination floor, or reserved door command bits during LC preparation.
- Verify that the LC treats repeated `0x12` hold frames as the same active robot maintenance request.
- Verify that the LC performs its preparation responsibilities:
  - Check lift readiness
  - Disable hall calls / normal passenger operation
  - Enter Robot Maintenance mode
  - Move/home the lift to the top service floor
  - Report ready for Jetson/robot control
- Verify that `0x11` reports the correct preparation states, such as:
  - Preparation Request Received
  - Checking Readiness
  - Entering Robot Maintenance Mode
  - Moving to Top Service Floor
  - Ready for Jetson/Robot Control
- Verify that Robot Maintenance Mode is ON when the lift enters Robot Maintenance mode.

- Verify that Lift Abnormal Mode is OFF before robot operation is allowed.
- Verify that Jetson Operation Permitted is ON before robot operation is allowed.
- Verify that the Jetson only proceeds when:

```
Automatic Service Session State = Ready for Jetson/Robot Control
Jetson Operation Permitted = 1
Robot Maintenance Mode = 1
Lift Abnormal Mode = 0
```

- Verify that the Jetson retries the `0x12` Start Automatic Service Preparation command until ACK or the expected `0x11` state transition is observed.

## Expected Result

The LC successfully handles automatic service preparation and reports `Ready for Jetson/Robot Control`. During this phase, the Jetson does not directly control lift movement or mode changes. The Jetson continues sending periodic `0x12` hold frames with `Automatic Service Request = 1` and `Automatic Service Request Type = 0x01`, while polling `0x11` for LC status.

---

# Phase 3: Lift Not Available Continuous Polling Test

## Objective

Verify that the Jetson handles the case where the lift is not available for automatic service.

## Scope

This phase focuses on the `Not Available` preparation path. The robot is not involved.

## Test Items

- Simulate or trigger a condition where the lift is not available for automatic service.
- Verify that the Jetson sends the `0x12` Start Automatic Service Preparation request.
- Verify that the LC reports `Automatic Service Session State = Not Available`.
- Verify that the Jetson continues polling LC status using `0x11`.
- Verify that the Jetson continues real-time status polling using `0x11` instead of using delayed retry polling.
- Verify that the Jetson does not start robot operation while the lift is not available.
- Verify that the Jetson proceeds only when the LC reports `Ready for Jetson/Robot Control`.
- Verify that the Jetson stops the workflow if the LC reports `Failed` or `Aborted`.
- Verify that while the lift is not available, the Jetson continues holding `Automatic Service Request = 1` and `Automatic Service Request Type = 0x01`.
- Verify that the Jetson continues polling `0x11` without using delayed 1-minute retry polling.

## Expected Result

The Jetson continuously polls LC status while the lift is not available and proceeds only when the LC reports `Ready for Jetson/Robot Control`. The Jetson uses continuous status polling.

---

# Phase 4: Jetson-Robot Integration Test

## Objective

Verify that the Jetson can communicate with and control the robot independently.

## Scope

This phase focuses only on communication between the Jetson and the robot. The Lift Controller is not involved.

## Test Items

- Verify that the Jetson can send commands to the robot.
- Verify that the robot can report its current status back to the Jetson.
- Verify basic robot movement commands.
- Verify that the Jetson can detect when the robot task is completed.
- Verify that the Jetson can detect robot error or abnormal states.
- Verify that the Jetson can stop the robot operation when required.
- Verify that the Jetson records robot fault details in its own logs.

## Expected Result

The Jetson can command the robot, receive robot status, detect task completion, detect robot faults, and stop robot operation when required.

---

# Phase 5: Manual Down Positioning Test

## Objective

Verify that the Jetson can position the lift at the required inspection height using Manual Down control, with `0x10` distance data as the primary positioning method and camera-based inspection point detection as the fallback method.

## Scope

This phase focuses on Jetson-controlled manual down movement after the LC has already reported `Ready for Jetson/Robot Control`.

## Test Items

- Verify that the LC has completed preparation and reports:

```
Automatic Service Session State = Ready for Jetson/Robot Control
```

- Verify that the Jetson reads Distance from Car Top to Current Floor Sill using `0x10`.
- Verify that the Jetson reads the current floor using `0x11`.
- Verify that the Jetson sends `0x12` with:

```
Manual Down = 1
Manual Up = 0
Reserved Door Command Bits = 0
Hall Call Disable = 1
Automatic Service Request = 1
Automatic Service Request Type = 0x01
```

- Verify that the Jetson continues polling `0x10` to monitor the sill distance.
- Verify that the Jetson continues polling `0x11` during manual down movement.
- Verify that manual down movement continues only while:
  - Automatic Service Session State = Ready for Jetson/Robot Control
  - Jetson Operation Permitted = ON
  - Robot Maintenance Mode = ON
  - Lift Abnormal Mode = OFF
- Verify that the primary positioning method uses `0x10` Distance from Car Top to Current Floor Sill.
- Verify that the Jetson turns Manual Down OFF when the `0x10` distance reaches the selected stopping condition.
- Verify that `Automatic Service Request` remains ON during Manual Down positioning.
- Verify that reserved door command bits remain OFF during Manual Down positioning.

- Verify that the fallback positioning method can use current floor information from `0x11` and camera-based detection of the defined inspection point.
- Verify that the Jetson turns Manual Down OFF when the camera detects the defined inspection point with sufficient confidence.
- Verify that the Jetson stops Manual Down and stops the inspection workflow if the selected positioning reference is not confirmed within the expected movement range, configured timeout, or safety limit.
- Verify that the Jetson sends `0x12` with:

```
Manual Down = 0
Manual Up = 0
Reserved Door Command Bits = 0
Hall Call Disable = 1
Automatic Service Request = 1
Automatic Service Request Type = 0x01
```

- Verify that the lift stops at the required inspection height before robot inspection begins.
- Verify that the Jetson does not allow robot inspection to start while the lift is still moving.

## Expected Result

The Jetson can move the lift downward using Manual Down control and stop the lift at the required inspection height using either the primary `0x10` distance-based method or the fallback camera-based inspection point method. Manual Down is safely turned OFF before robot inspection begins.

---

# Phase 6: Robot Inspection Hold Test

## Objective

Verify that robot inspection is only performed when the lift is stationary and the LC still permits Jetson/robot control.

## Scope

This phase focuses on the coordination between lift hold state and robot inspection.

## Test Items

- Verify that the robot inspection starts only after:
  - Lift is stationary
  - Manual Down = OFF
  - Manual Up = OFF
  - Automatic Service Session State = Ready for Jetson/Robot Control
  - Jetson Operation Permitted = ON
  - Robot Maintenance Mode = ON
  - Lift Abnormal Mode = OFF
- Verify that the Jetson continues polling `0x11` during robot inspection.
- Verify that during robot inspection hold state, the Jetson sends / maintains:
  - Manual Down = 0
  - Manual Up = 0
  - Reserved Door Command Bits = 0
  - Hall Call Disable = 1
  - Automatic Service Request = 1
  - Automatic Service Request Type = 0x01
- Verify that the Jetson does not send new lift movement commands during robot inspection.
- Verify that the LC does not execute independent lift movement or door movement during robot inspection.
- Verify that the Jetson stops robot operation if Jetson Operation Permitted turns OFF, Robot Maintenance Mode turns OFF, or Lift Abnormal Mode turns ON.
- Verify that the robot reports inspection completion to the Jetson.
- Verify that the Jetson does not move to the next service floor until the robot inspection is complete.

## Expected Result

Robot inspection only occurs when the lift is stationary, `Automatic Service Session State = Ready for Jetson/Robot Control`, `Jetson Operation Permitted = ON`, `Robot`

`Maintenance Mode = ON`, and `Lift Abnormal Mode = OFF`. Lift movement and robot operation do not happen at the same time.

---

# Phase 7: Multi-Floor Service Sequence Test

## Objective

Verify that the Jetson can repeat the manual down positioning and robot inspection sequence across the selected service floors.

## Scope

This phase tests the repeated inspection sequence from the top service floor downward.

## Test Items

- Verify that the Jetson reads the service floor configuration from `0x10`.
- Verify that the Jetson identifies valid service floors.
- Verify that non-service floors are skipped.
- Verify that the Jetson moves downward from the top service floor toward lower service floors.
- Verify that the Jetson only stops when the selected positioning reference corresponds to the intended inspection point for the current service floor.
- Verify that each valid service floor follows this sequence:
  1. Poll `0x11`
  2. Confirm `Ready for Jetson/Robot Control`
  3. Confirm Jetson Operation Permitted is ON, Robot Maintenance Mode is ON, and Lift Abnormal Mode is OFF.
  4. Send `0x12` with Manual Down ON, `Automatic Service Request = 1`, and `Automatic Service Request Type = 0x01`
  5. Monitor the selected positioning reference using `0x10` Distance from Car Top to Current Floor Sill and/or camera-based inspection point detection.
  6. Send Manual Down OFF while keeping `Automatic Service Request = 1` and `Automatic Service Request Type = 0x01`.
  7. Hold lift stationary
  8. Run robot inspection
  9. Continue to the next lower service floor
- Verify that the system does not attempt to service below the 2nd floor.
- Verify that `Automatic Service Request` remains ON across all serviced floors until the final inspection point is completed.

## Expected Result

The Jetson correctly repeats the inspection workflow for each valid service floor, skips non-service floors, stops only when the selected positioning reference matches the intended

inspection point, logs each inspection result, and stops after the lowest required service point is completed.

---

# Phase 8: LC Automatic Service Completion Test

## Objective

Verify that the LC can complete the automatic service session and return the lift to normal operation after receiving the Jetson's completion request.

## Scope

This phase focuses on the Jetson-LC completion flow after the final inspection point is completed. The robot is no longer operating.

## Test Items

- Verify that after the final inspection point is completed, the Jetson sends `0x12` with:

```
Manual Down = 0
Manual Up = 0
Reserved Door Command Bits = 0
Hall Call Disable = 0
Automatic Service Request = 0
Automatic Service Request Type = 0x00
```

- Verify that the LC acknowledges the `0x12` command.
- Verify that the Jetson does not send Manual Up, Manual Down, destination floor, or door commands while LC completion is in progress.
- Verify that the drop from `Automatic Service Request = 1` to `Automatic Service Request = 0` is treated as the completion / return-to-normal request.
- Verify that the Jetson retries the request-cleared command until ACK or the expected completion-related `0x11` state transition is observed.
- Verify that after clearing `Automatic Service Request` to OFF, the Jetson only polls `0x11` during LC completion.
- LC handles the completion responsibilities:
  - Restore Auto mode from the current lift position
  - Level the lift as required
  - Re-enable hall calls / normal passenger operation
  - Report service complete
- Verify that `0x11` reports the correct completion states, such as:
  - Completion Request Received
  - Returning to Auto Mode
  - Service Complete / Normal Operation Restored
- Verify that the final LC status is:

```
Automatic Service Session State = Service Complete / Normal Operation
Restored
Robot Maintenance Mode = 0
Lift Abnormal Mode = 0
```

## Expected Result

The LC successfully completes the automatic service session, restores Auto mode, re-enables normal operation, and reports service completion.

---

# Phase 9: Full Flow Test Without 3am Trigger

## Objective

Verify the complete maintenance workflow manually without using the scheduled 3am trigger.

## Scope

This phase tests the full interaction between the Jetson, LC, camera/vision module, and robot, but the sequence is started manually instead of automatically at 3am.

## Test Items

- Manually start the maintenance workflow from the Jetson.
- Verify that the Jetson starts an automatic service session log.
- Verify that the Jetson sends `0x12` Start Automatic Service Preparation.
- Verify that the LC handles preparation internally.
- Verify that during LC preparation, the Jetson continues sending periodic `0x12` hold frames with `Automatic Service Request = 1` and `Automatic Service Request Type = 0x01`, while polling `0x11`.
- Verify that the LC reports `Ready for Jetson/Robot Control`.
- Verify that the Jetson checks robot readiness before starting inspection.
- Verify that the Jetson uses `0x10` Distance from Car Top to Current Floor Sill as the primary positioning method.
- Verify that the Jetson can use current floor information from `0x11` and camera-based inspection point detection as the fallback positioning method.
- Verify that Manual Down is turned OFF before robot inspection begins.
- Verify that robot inspection starts only when the lift is stationary.
- Verify that lift movement and robot movement do not occur at the same time.
- Verify that the sequence repeats correctly for lower service floors.
- Verify that non-service floors are skipped.
- Verify that each service floor result is recorded in the Jetson logs.
- Verify that after the final inspection point, the Jetson clears `Automatic Service Request = 0` with `Automatic Service Request Type = 0x00` to request return-to-normal handling.
- Verify that `Automatic Service Request = 1` and `Automatic Service Request Type = 0x01` are held throughout preparation, manual down positioning, and robot inspection.
- Verify that the LC handles completion internally.
- Verify that the Jetson only polls `0x11` during LC completion.
- Verify that the LC reports `Service Complete / Normal Operation Restored`.
- Verify that the Jetson closes the automatic service session log after completion.

### Expected Result

The full maintenance sequence can be completed manually without relying on the scheduled trigger. The LC/Jetson responsibility split, primary distance positioning, fallback camera positioning, robot inspection flow, and session logging are verified from preparation to completion.

## Phase 10: Full Flow Test With 3am Trigger

### Objective

Verify the complete automatic nightly maintenance workflow under the intended scheduled operating condition.

### Scope

This phase tests the full system, including the 3am trigger, Jetson-LC communication, LC automatic service preparation, lift positioning, robot operation, abnormal handling, maximum servicing duration monitoring, and LC automatic service completion.

### Test Items

- Verify that the Jetson starts the maintenance sequence automatically at `03:00`.
- Verify that the Jetson starts the maximum servicing duration timer.
- Verify that the Jetson starts an automatic service session log.
- Verify that the LC checks whether the lift is available for automatic service.
- Verify that the Jetson continues polling `0x11` if the lift is not available.
- Verify that the LC disables hall calls, enters Robot Maintenance mode, and moves/homes the lift to the top service floor.
- Verify that the LC reports `Ready for Jetson/Robot Control`.
- Verify that periodic `0x12` hold frames are sent while robot maintenance operation is active.
- Verify that robot maintenance operation ends only after `Automatic Service Request` is cleared to OFF.
- Verify that the Jetson confirms robot readiness.
- Verify that the Jetson uses `0x10` Distance from Car Top to Current Floor Sill as the primary positioning method.
- Verify that the Jetson can use current floor information from `0x11` and camera-based inspection point detection as the fallback positioning method.
- Verify that the robot performs inspection only when the lift is stationary at the inspection height.
- Verify that the sequence repeats for each valid service floor.
- Verify that non-service floors are skipped.
- Verify that each service floor result is recorded in the Jetson logs.

- Verify that the Jetson monitors the total servicing duration throughout the workflow.
- Verify that the final inspection point may be at any service floor depending on the inspection sequence.
- Verify that after the final inspection point, the Jetson clears `Automatic Service Request = 0` with `Automatic Service Request Type = 0x00` to request return-to-normal handling.
- Verify that the LC restores Auto mode from the current lift position, levels the lift as required, and re-enables normal operation.
- Verify that robot inspection only starts when Jetson Operation Permitted is ON, Robot Maintenance Mode is ON, and Lift Abnormal Mode is OFF.
- Verify that the LC reports `Service Complete / Normal Operation Restored`.
- Verify that the Jetson stops the maximum servicing duration timer and closes the automatic service session log.

### Expected Result

The complete automatic nightly maintenance workflow runs successfully from the 3am trigger to final return-to-normal operation, with correct coordination between the Jetson, LC, camera/vision module, and robot. The system records the automatic service session and monitors the maximum servicing duration throughout the workflow.

---

## Phase 11: Abnormal Handling Test

### Objective

Verify that the system responds safely when abnormal conditions occur during automatic service.

### Scope

This phase tests abnormal and override scenarios for the Jetson, LC, and robot.

### Test Items

#### LC No Response

- Simulate LC no response.
- Verify that the Jetson retries communication.
- Verify that timeout handling is triggered if communication is not restored.
- Verify that an alert is raised if communication is not restored within the configured time.

#### LC No Response During Manual Down

- Simulate LC no response while Manual Down is active.

- Verify that the Jetson immediately attempts to clear Manual Up and Manual Down.
- Verify that the Jetson continues retrying communication.
- Verify that an alert is raised if communication is not restored.

## Lift Abnormal Mode

- Simulate Lift Abnormal Mode during automatic service.
- Verify that the Jetson immediately stops robot operation if active.
- Verify that the Jetson clears Manual Up and Manual Down if manual movement is active.
- Verify that the Jetson stops the maintenance workflow.
- Verify that the system waits for recovery or manual intervention.

## LC State Not Ready During Inspection

- Simulate the LC state changing away from `Ready for Jetson/Robot Control` during manual down positioning or robot inspection.
- Verify that the Jetson immediately stops robot operation.
- Verify that the Jetson stops the inspection workflow.
- Verify that the Jetson clears Manual Down if manual movement is active.
- Verify that the Jetson reads the Automatic Service Abnormal Reason if available.

This includes:

- Automatic Service Session State is no longer Ready for Jetson/Robot Control
- Jetson Operation Permitted turns OFF
- Robot Maintenance Mode turns OFF
- Lift Abnormal Mode turns ON

## Technician Manual Intervention

- Simulate technician/manual takeover during robot inspection.
- Verify that the LC reports:

```
Automatic Service Session State = Aborted
Automatic Service Abnormal Reason = Manual Technician Intervention / Manual
Service Takeover
```

- Verify that the Jetson stops robot operation.
- Verify that the Jetson stops sending lift movement commands.
- Verify that automatic servicing does not resume automatically.
- Verify that manual recovery is required.

## Robot Fault

- Simulate robot fault during inspection.
- Verify that the Jetson detects the robot fault.
- Verify that the Jetson stops robot operation.
- Verify that the Jetson stops the inspection workflow.
- Verify that the Jetson clears active lift movement commands if manual movement is active.
- Verify that the Jetson sends `0x12` Abort Robot Maintenance Operation with:

```
Automatic Service Request = 1
Automatic Service Request Type = 0x02
Manual Up = 0
Manual Down = 0
Reserved Door Command Bits = 0
```

- Verify that the Jetson continues sending the abort request until the LC acknowledges it or `0x11` reports `Automatic Service Session State = Aborted`.
- Verify that the LC reports:

```
Automatic Service Session State = Aborted
Automatic Service Abnormal Reason = Abort Requested by Jetson
```

- Verify that the detailed robot fault reason is stored in the Jetson logs.
- Verify that operator or technician recovery is required.

### Jetson Communication Lost

- Simulate Jetson communication loss during automatic service.
- Verify that the LC handles Jetson communication loss according to the final engineering decision.
- Verify that the LC reports Jetson communication loss if supported:

```
Automatic Service Abnormal Reason = Jetson Communication Lost
```

## Expected Result

The system handles abnormal situations safely by stopping robot operation, stopping or preventing lift movement when required, reporting the correct session state and abnormal reason, and requiring manual recovery where appropriate.

---

# Phase 12: Manual Down Safety Limit Test

## Objective

Verify that Manual Down movement cannot continue indefinitely if the Jetson fails to clear the Manual Down command.

## Scope

This phase focuses on LC-side safety limits during Jetson-controlled manual down movement.

## Test Items

- Verify that Manual Down is only active while moving toward the selected stopping condition for the current inspection point.
- Verify that the Jetson turns Manual Down OFF immediately when the selected stopping condition is reached.
- Verify that Manual Down is only active while `Automatic Service Request = 1` and `Automatic Service Request Type = 0x01`.
- Verify that Manual Down is cleared if Robot Maintenance Mode turns OFF.
- Verify that Reserved Door Command Bits remain OFF during manual down movement.
- Simulate the Jetson failing to clear Manual Down.
- Verify LC-side safety handling for maximum manual down duration.
- Verify LC-side safety handling for maximum movement distance.
- Verify LC-side safety handling if distance data is invalid or not changing.
- Verify LC-side safety handling if Jetson communication is lost.
- Verify LC-side / Jetson-side safety handling if the camera-based inspection point is not detected within the expected movement range or configured timeout.
- Verify that Manual Down is cleared if Lift Abnormal Mode turns ON.
- Verify that Manual Down is cleared if Jetson Operation Permitted turns OFF.

## Expected Result

Manual Down movement is safely bounded. The lift does not continue moving indefinitely if the Jetson fails to clear Manual Down or if communication/data becomes invalid.

---

# Phase 13: Maximum Servicing Duration Exceeded Test

## Objective

Verify that the system stops the automatic maintenance workflow and returns the lift to normal passenger operation when the configured maximum servicing duration is exceeded.

## Scope

This phase tests the maximum servicing duration rule. The initial proposed maximum duration is `1 hour`, subject to further discussion and confirmation.

## Test Items

- Configure or simulate the maximum servicing duration being exceeded.
- Verify that the maximum servicing duration is measured from the time the Jetson sends `0x12` Start Automatic Service Preparation.
- Verify that the Jetson detects when the configured maximum servicing duration has been exceeded.
- Verify that the Jetson stops the current robot operation if one is active.
- Verify that the Jetson clears Manual Up and Manual Down commands if manual movement is active.
- Verify that the Jetson stops further robot inspection for the remaining service floors.
- Verify that the Jetson records the automatic service session result in its logs as incomplete due to maximum servicing duration exceeded.
- Verify that the Jetson clears `Automatic Service Request = 0` with `Automatic Service Request Type = 0x00` to request LC return-to-normal handling.
- Verify that the Jetson clears Manual Up and Manual Down before or together with clearing `Automatic Service Request` to OFF.
- Verify that the LC detects the request-cleared state and begins return-to-normal handling.
- Verify that the Jetson continues polling LC status using `0x11`.
- Verify that the LC handles the return-to-normal sequence internally.
- Verify that the LC restores Auto mode from the current lift position, levels the lift as required, and re-enables normal passenger operation.
- Verify that the LC reports:

```
Automatic Service Session State = Service Complete / Normal Operation
Restored
```

- Verify that the Jetson closes the automatic service session log after the lift returns to normal operation.
- Verify that if the LC cannot safely return to normal operation due to an active safety fault, interlock condition, Lift Abnormal Mode or other LC-side abnormal condition, LC safety logic takes priority and the Jetson keeps the robot stopped while polling `0x11`.

## Expected Result

When the maximum servicing duration is exceeded, the Jetson stops the remaining maintenance workflow, logs the session as incomplete, and requests the LC to return the lift

to normal passenger operation. The lift returns to normal operation unless prevented by LC-side safety logic.